

Encryption Technologies

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Cryptography

- "Cryptography" comes from the Greek words kryptos, meaning "concealed, hidden, veiled, secret, or mysterious," and graphia, meaning "writing"; thus, cryptography is "the art of secret writing."
- Cryptography is the practice of concealing information by converting plaintext (readable format) into ciphertext (unreadable format) using a key or encryption scheme.
- It is the process of converting data into a scrambled code that is encrypted and sent across a private or public network.
- **Encryption** is the method by which information is converted into secret code that hides the information's true meaning
- Encrypted messages can, at times, be decrypted by **cryptanalysis** (code breaking), even though modern encryption techniques are virtually unbreakable.

Objectives of Cryptography

- **Confidentiality**: Assurance that the information is accessible only to those authorized to access it.
- **Integrity:** Trustworthiness of data or resources in terms of preventing improper and unauthorized changes.
- **Authentication:** Assurance that the communication, document, or data is genuine.
- Nonrepudiation: Guarantee that the sender of a message cannot later deny having sent the message and that the recipient cannot deny having received the message.

Cryptography Terminology

- **Plain text** – Message in its original form (readable text)
- **Cipher text** – Message in Coded Form
- **Encryption** – Transforming Plain text to Cipher text
- **Decryption** – transforming cipher text to plain text
- **Encryption algorithm** – Method used for encryption

Cryptography Process

- The cryptography process involves transforming a Plaintext **(readable format)** into a cipher text using encryption algorithms.
- On reaching the destination, is decrypt cipher text into readable plaintext.



Types of Cryptography

- Cryptography is categorized into two types according to the number of keys employed for encryption and decryption:
  - Symmetric encryption
  - Asymmetric Encryption

Symmetric encryption

- Symmetric encryption use the same pre-shared key (secret key) to encrypt and decrypt data.
- The sender uses a key to encrypt the plaintext and sends the resultant cipher text to the recipient, who uses the same key (used for encryption) to decrypt the cipher text into plaintext.
- Symmetric encryption is also known as secret-key cryptography, as it uses only one secret key to encrypt and decrypt the data.

Symmetric encryption



Symmetric encryption

- Symmetric encryption comes in two forms, these are:
  - Block cipher
  - Stream ciphers

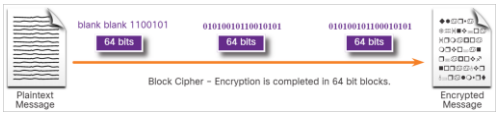
Symmetric Encryption (Contd.)

Block Ciphers

- A block cipher is a method of encrypting data in blocks to produce cipher text using a cryptographic key and algorithm.
- The block cipher processes fixed-size blocks simultaneously
- Block ciphers transform a fixed-length block of plaintext into a common block of ciphertext of 64 or 128 bits.
- Common block ciphers include Data Encryption Standard(DES) with a 64-bit block size and Advanced Encryption Standard (AES) with a 128-bit block size.

Symmetric encryption

Block Ciphers

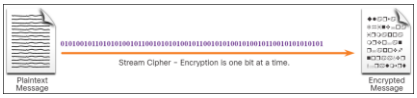


Symmetric Encryption (Contd.)

Stream Ciphers

- Stream ciphers encrypt plaintext one byte or one bit at a time.
- Stream ciphers are basically a block cipher with a block size of one byte or bit.
- Stream ciphers are typically faster than block ciphers because data is continuously encrypted.
- Examples include RC4 and A5 which is used to encrypt Global System for Mobile communication(GSM)cell phone communications.

Symmetric encryption



Uses of symmetric encryption

1. **Data at rest:** Symmetric cryptography is also widely used to secure data stored in hard drives, archives, computers, flash drives, and laptops.
2. **Banking sector** – used Symmetric encryption to encrypt bulk data within the banking sector
3. **Messaging apps** like WhatsApp and signal uses end to end encryption to safeguard the privacy and confidentiality of communications on the platforms and also in authenticating users.

Symmetric encryption

- This type of cryptography works well when you are communicating with only a few people.
- Because the sender and receiver must share the key before sending any messages, this technique is of limited use for the Internet, where individuals who have not had prior contact frequently require a secure means of communication.
- The solution to this problem is asymmetric encryption (public-key cryptography).

Asymmetric Encryption

- An asymmetric-key system is an encryption method that uses a key pair comprising a public key available to anyone and a private key held only by the key owner, which helps to provide confidentiality, integrity, authentication, and nonrepudiation in data management.
- The concept of asymmetric encryption (also known as public-key cryptography) was introduced to solve key-management problems.
- Asymmetric encryption involves both a public key and a private key.

Asymmetric Encryption

Asymmetric encryption uses the following sequence to send a message:

1. An individual finds the public key of the person he or she wants to contact in a directory.
2. This public key is used to encrypt a message that is then sent to the intended recipient.
3. The receiver uses the private key to decrypt the message and reads it.

Asymmetric Encryption

- No one but the holder of the private key can decrypt a message encrypted with the corresponding public key.
- This increases the security of the information because all communications involve only public keys; the message sender never transmits or shares the private keys.
- The sender must link public keys with usernames in a secure manner to ensure that individuals claiming to be the intended recipient do not intercept the information.



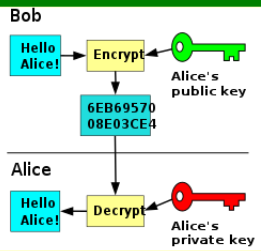
Asymmetric encryption

- Common examples of asymmetric encryption algorithms
  - Digital Signature Standard (DSS) and Digital Signature Algorithm (DSA)
  - Diffie-Hellman (DH)
  - Elliptic curve techniques

Asymmetric Encryption

- Asymmetric algorithms are used to provide confidentiality without pre-sharing a password.
- The confidentiality objective of asymmetric algorithms is initiated when the encryption process is started with the public key.
- The process can be summarized using the formula: **Public Key (Encrypt) + Private Key (Decrypt) = Confidentiality**
- When the public key is used to encrypt data, the private key must be used to decrypt data.
- Only one host has the private key; therefore, confidentiality is achieved.

Asymmetric Encryption



Discussion

Group 1: What do you think are the Strengths and weaknesses of asymmetric encryption?

Group 2 : What do you think are the Strengths and weaknesses of symmetric encryption?

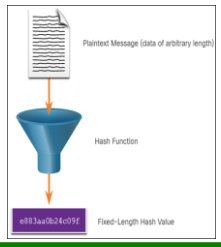
Strengths and weaknesses of asymmetric and symmetric encryption

- Take note of the weaknesses and strengths discussed

Hash Functions

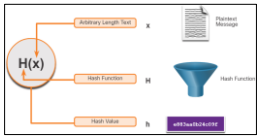
- Hashing is based on a one-way mathematical function that is relatively easy to compute, but significantly harder to reverse.
- Hashes are used to verify and ensure data integrity.
- A hash function takes a variable block of binary data, called the message, and produces a fixed-length, condensed representation, called the hash.
- The resulting hash is also sometimes called the message digest, digest, or digital fingerprint.
- With hash functions, it is computationally infeasible for two different sets of data to come up with the same hash output.
- Every time the data is changed or altered, the hash value also changes – used in checking integrity of data

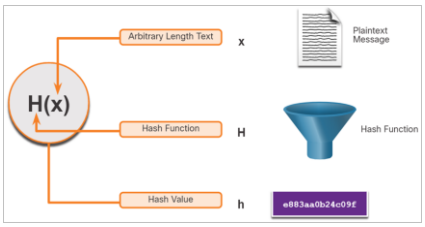
How Hashing Works



Hash Operation

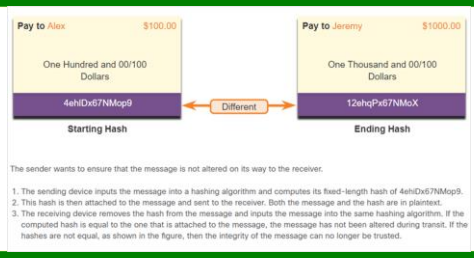
- Mathematically, the equation ***h= H(x)*** is used to explain how a hash algorithm operates.
- As shown in the figure, a hash function H takes an input x and returns a fixed-size string hash value h.
- A cryptographic hash function should have the following properties:
  - The input can be any length.
  - The output has a fixed length.
  - H(x) is relatively easy to compute for given x.
  - H(x) is one way and not reversible.
  - H(x) is collision free, meaning that two different input values will result in different hash values.





Hash Functions

- Hash functions are used to ensure the integrity of a message either accidentally or intentionally.
- In the figure, the sender is sending a $100 money transfer to Alex. The sender wants to ensure that the message is not altered on its way to the receiver.



MD5 and SHA

There are four well-known hash functions:

- **MD5 with 128-bit digest** - A one-way function that produces a 128-bit hashed message. MD5 is a legacy algorithm.
- **SHA-1 -** Very similar to the MD5 hash functions. SHA-1 creates a 160-bit hashed message and is slightly slower than MD5.
- **SHA-2 -** If you are using SHA-2, then SHA-256, SHA-384, and SHA-512 algorithms should be used.
  - **SHA-3 -** Next-generation algorithms and should be used whenever possible.

MD5 and SHA

- While hashing can be used to detect accidental changes, it cannot be used to guard against deliberate changes that are made by a threat actor.
- There is no unique identifying information from the sender in the hashing procedure.
- This means that anyone can compute a hash for any data, as long as they have the correct hash function.
- Therefore, hashing is vulnerable to man-in-the-middle attacks and does not provide security to transmitted data. To provide integrity and origin authentication, something more is required.

Characteristics of a Hashing algorithm

- A hashing algorithm is considered secure if it has these characteristics:
  - **Fixed size**. A digest of a short set of data should produce the same size as a digest of a long set of data. For example, a digest of the single letter a is 86be7afa339d0fc7cfc785e72f578d33, while a digest of 1 million occurrences of the letter a is 4a7f5723f954eba1216c9d8f6320431f, the same length.
  - **Unique.** Two different sets of data cannot produce the same digest, which is known as a collision. Changing a single letter in one data set should produce an entirely different digest. For example, a digest of Sunday is 0d716e73a2a7910bd4ae63407056d79b while a digest of sunday (lowercase s) is 3464eb71bd7a4377967a30da798a1b54.
  - **Original.** It should be impossible to produce a data set that has a desired or predefined hash.
  - **Secure.** The resulting hash cannot be reversed in order to determine the original plaintext.

Applications of Hash Functions

1. **Password Storage**
   - Hash functions provide protection to password storage.
   - Instead of storing password in clear, mostly all logon processes store the hash values of passwords in the file.
   - Example: Microsoft stores password as hashes, most websites also stores password as hashes.
2. **Data Integrity Check**
   - Data integrity check is a most common application of the hash functions. It is used to generate the checksums on data files.
   - This application provides assurance to the user about correctness of the data.
   - Example: sites that run downloads

Encryption vs Hashing

What are the differences between Hashing and Encryption

LAB 1 – To do at home

Creating Hashes in Windows
Activity(15 Min)

Simple Cryptography – Caesar Cipher

- Caesar Cipher is the earliest ciphers developed by Caeser
- The original one, each letter in a message is substituted by another that is 3 letters away i.e. a becomes D
- Also known as shift cipher
- For Example : Attack at Dawn

Caesar Cipher

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Diffie-Hellman

- Diffie-Hellman key exchange is a protocol that two people can use over a shared network connection to share a key they want to use for encryption i.e in symmetric key encryption.
- How do you share key encryption with someone before you both know the key to use for encryption – Use differ hellman.

Video

- Video explaining how diffie Hellman works

Digital Signature

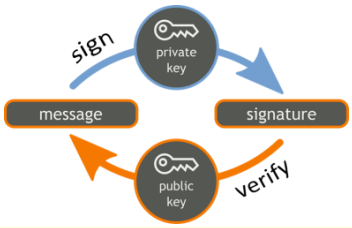
- A digital signature is a small amount of data that was created using some private key, and there is a public key that can be used to verify that the signature was really generated using the corresponding private key.
- Originates from asymmetric cryptography
- Digital signature is a cryptographic value that is calculated from the data and a secret key known only by the signer.
- The algorithm used to generate the signature must be such that without knowing the private key it is not possible to create a signature that would verify as valid.
- Digital signatures are used to verify that a message really comes from the claimed sender (assuming only the sender knows the private key corresponding to the public key). This is called (data origin) authentication.

Digital Signature

- They can also be used to timestamp documents: a trusted party signs the document and its timestamp with his/her private key, thus testifying that the document existed at the stated time.
- Digital signatures can also be used to certify that a public key belongs to a particular entity.

Digital Signatures: How they work

A message is signed by a private key and the signature is verified by the corresponding public key



Digital Signatures: How they work

Messages are signed by the sender using a private key (signing key).

Typically the input message is hashed and then the signature is calculated by the signing algorithm.

Most signature algorithms perform some calculation with the message hash + the signing key in a way that the result cannot be calculated without the signing key.

- The result from message signing is the digital signature (one or more integers):
- **signMsg(msg, privKey)**

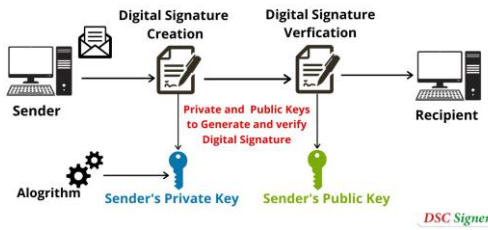
Message signatures are verified by the corresponding public key (verification key).

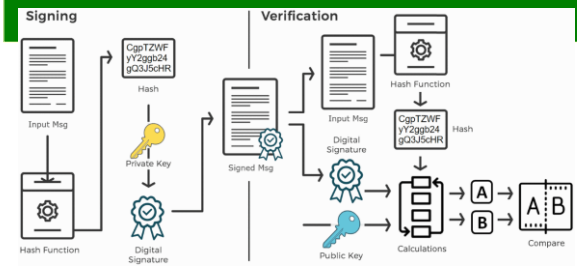
Typically the signed message is hashed and some calculation is performed by the signature algorithm using the message hash + the public key.

The result from signing is a boolean value (valid or invalid signature):

verifyMsgSignature(msg, signature, pubKey) valid / invalid

How Digital Signature Works





Importance of Digital signatures

1. **Message authentication** − When the verifier validates the digital signature using public key of a sender, he is assured that signature has been created only by sender who possess the corresponding secret private key and no one else.
2. **Data Integrity** − In case an attacker has access to the data and modifies it, the digital signature verification at receiver end fails. The hash of modified data and the output provided by the verification algorithm will not match. Hence, receiver can safely deny the message assuming that data integrity has been breached.
3. **Non-repudiation** − Since it is assumed that only the signer has the knowledge of the signature key, he can only create unique signature on a given data. Thus the receiver can present data and the digital signature to a third party as evidence if any dispute arises in the future.

Implementing encryption in networks

- The implementation of encryption in packet-switched networks must ensure that essential addressing information can be accessed by the relevant network devices such as switches, bridges and routers.
- Encryption is broadly termed link layer encryption or end-to-end encryption depending on whether it is applied and re-applied at each end of each link in a communication path, or whether it is applied over the whole path between end systems.
- It is useful to identify the various implementations of encryption with the appropriate OSI layer
- End-to-end encryption is implemented at or above layer 3, the network layer of the OSI reference model, while link layer encryption is applied at the data link and physical layers.
- When encryption is applied at the transport or network layers, end-to-end refers to hosts identified by IP (internet protocol) addresses and, in the case of TCP (transmission control protocol) connections, port numbers. In the context of application layer encryption, however, end-to-end is more correctly interpreted as **process-to-process**

Implementing encryption in networks

- Confidentiality between two communicating nodes is achieved by using an appropriate encryption scheme: data is encrypted at the sending node and decrypted at the receiving node.
- Encryption will also protect the traffic between the two nodes from eavesdropping to some extent.
- However, for encryption to be used effectively in networks, it is necessary to define what will be encrypted, where this takes place in the network, and the layers that are involved in a reference model.

Cryptanalysis

- Cryptanalysis is used as a method of decrypting a cipher text into plaintext.
- An unauthorized person tries to decrypt the message by eavesdropping on the unsecured channel. It is also known as code-breaking.
- This person is not bound by any rules i.e a black hat hacker

Network security protocols

- Network security protocols are a type network protocol that ensures the security and integrity of data in transit over a network connection.
- Network security protocols define the processes and methodology to secure network data from any illegitimate attempt to review or extract the contents of data.
- Network security protocols generally implement cryptography and encryption techniques to secure the data so that it can only be decrypted with a special algorithm, logical key, mathematical formula and/or a combination of all of them.
- Some of the popular network security protocols include Secure File Transfer Protocol (SFTP), Secure Hypertext Transfer Protocol (HTTPS) and Secure Socket Layer (SSL).